

Disk

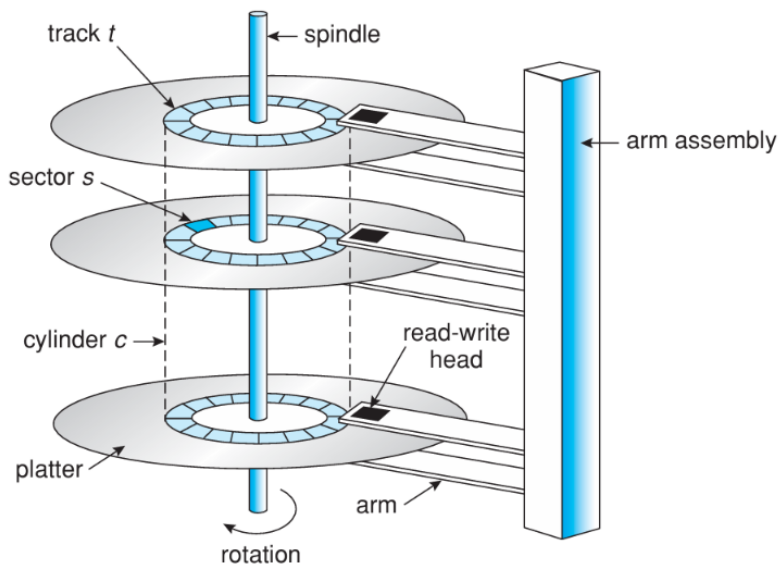
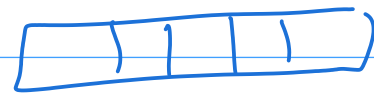


Figure 10.1 Moving-head disk mechanism.

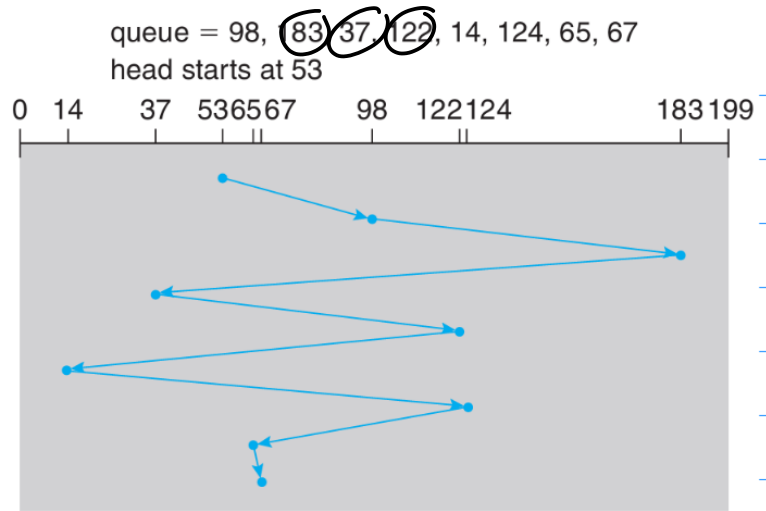


Figure 10.4 FCFS disk scheduling.

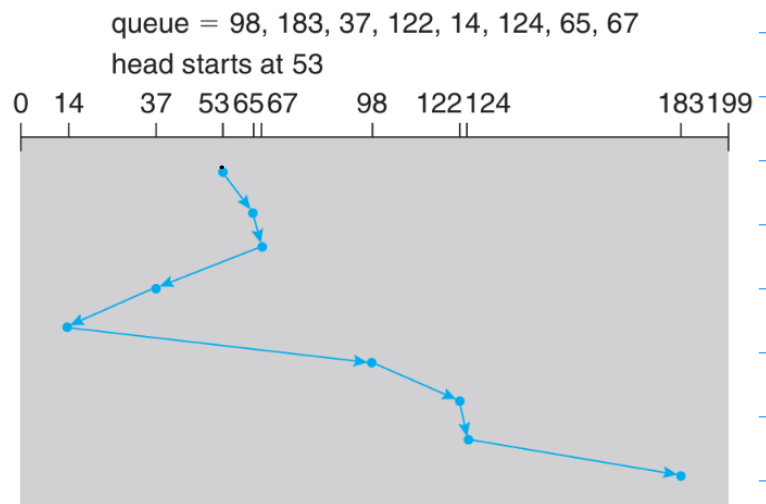


Figure 10.5 SSTF disk scheduling.

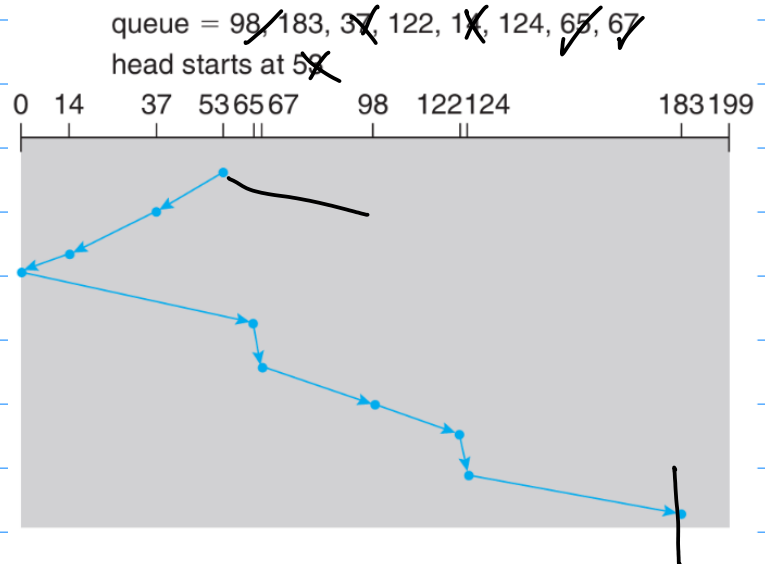


Figure 10.6 SCAN disk scheduling.

look
C-look

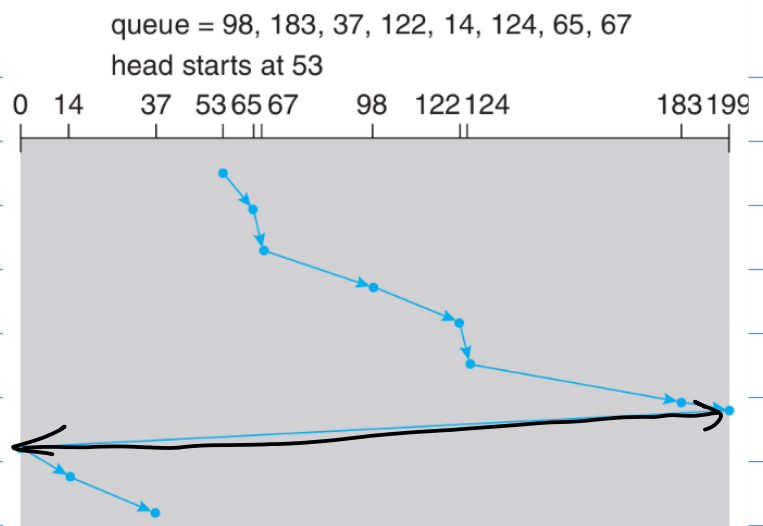


Figure 10.7 C-SCAN disk scheduling.

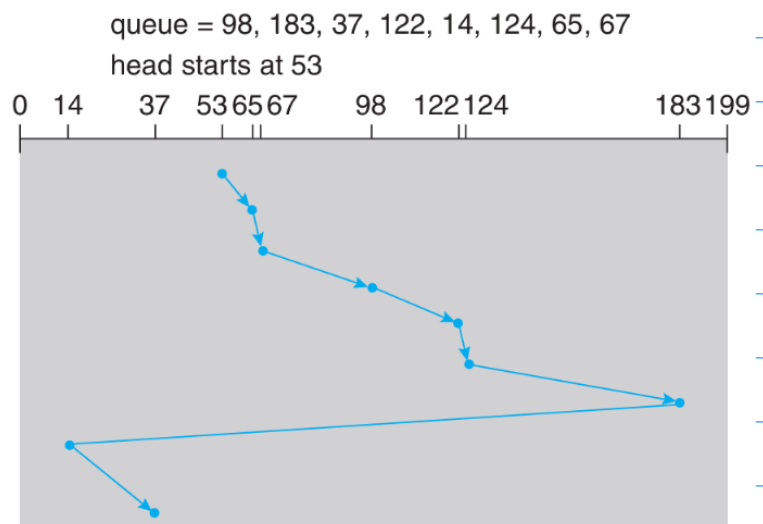


Figure 10.8 C-LOOK disk scheduling.

10.11 Suppose that a disk drive has 5,000 cylinders, numbered 0 to 4,999. The drive is currently serving a request at cylinder 2,150, and the previous request was at cylinder 1,805. The queue of pending requests, in FIFO order, is:

2,069, 1,212, 2,296, 2,800, 544, 1,618, 356, 1,523, 4,965, 3681

Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms?

- a. FCFS
- b. SSTF
- c. SCAN
- d. LOOK
- e. C-SCAN
- f. C-LOOK

2150

2,069, 1,212, 2,296, 2,800, 544, 1,618, 356, 1,523, 4,965, 3681

2150, 2296, 2800, 3681, 4965, 4999
0, 356, 544, 1212, 1523, 1618
2069

F C F S

2150, 2069, 1212, 2296, 2800, 544, 1618, 356, 1523, 4965, 3681

13011

S S T F

2150, 2069, 2296, 2800, 3681, 4965, 1618, 1523, 1212, 544, 356.

7586

S c a n

2150, 2296, 2800, 3681, 4965, 4999, 2069, 1618, 1523, 1212, 544, 356.

7492

Look

2150, 2296, 2800, 3681, 4965, 2069, 1618, 1523, 1212, 544, 356

7424

C-Scan

2150, 2296, 2800, 3681, 4965, 4999, 0, 356, 544, 1212, 1523, 1618, 2069

4917

C-look

2150, 2296, 2800, 3681, 4965, 356, 544, 1212, 1523, 1618, 2069

4528